

Does Management Uncertainty in Earnings Calls Predict Short-Term Stock Volatility?

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Abstract

We investigate whether uncertainty-related language used by management during quarterly earnings calls predicts short-term stock price volatility. Using over 100,000 earnings-call transcripts scraped from Koyfin (2021–2026), we construct a sentence-level uncertainty measure by fine-tuning a FinBERT model on silver labels derived from the Loughran and McDonald (2011) Uncertainty word list. The fine-tuned classifier achieves an F1 score of 0.82 on a held-out test set. In panel regressions with sector and year-quarter fixed effects and firm-clustered standard errors, we find that a 1 percentage point increase in the model-predicted uncertainty score is associated with approximately 49 basis points higher annualised post-call realised volatility ($p = 0.003$). The effect is robust to alternative volatility windows, an alternative uncertainty measure, and firm-size sub-samples. Notably, uncertainty expressed during prepared remarks is almost six times more predictive than uncertainty arising in the Q&A session.

1 Introduction

Quarterly earnings calls are among the most closely monitored corporate events in financial markets. Beyond the hard numbers reported in earnings releases, the language used by management conveys qualitative, “soft” information that investors use to update expectations (Kearney and Liu, 2014). A growing literature shows that linguistic features of corporate communications — sentiment, tone, complexity — contain predictive content for asset prices and volatility (Loughran and McDonald, 2011; Huang et al., 2014).

In this paper, we ask: **does management uncertainty in earnings calls predict short-term stock volatility?** Our hypothesis is that firms whose management uses more uncertainty-laden language experience higher realised volatility in the trading days following the call, reflecting increased investor disagreement about the firm’s outlook. To test this, we fine-tune FinBERT (Araci, 2019) — a domain-specific transformer — to classify individual sentences as high or low uncertainty, rather than relying on simple dictionary word counts. Our pipeline processes over 100,000 transcripts spanning five years and links the resulting uncertainty scores to post-call volatility through panel regressions with sector and year-quarter fixed effects, firm-clustered standard errors, and multiple robustness checks.

Our main finding is that a 1 percentage point increase in the uncertainty score is associated with approximately 49 basis points higher annualised post-call volatility ($p = 0.003$). The effect is concentrated in prepared remarks rather than Q&A, robust across alternative volatility windows and firm sizes, and strongest in Health Care and Financials.

2 Data

2.1 Earnings-Call Transcripts

We collect approximately 130,000 quarterly earnings-call transcripts from the Koyfin platform, covering Q1 2021 to Q1 2026. To retrieve the transcripts, we do not rely on standard webpage scraping, as this approach would be too slow and inefficient at scale. Instead, Selenium is only used to open a headless browser, log into Koyfin, and recover the authentication token from the session. Once authenticated, the scraper accesses Koyfin’s internal transcript endpoints through JavaScript `fetch()` calls executed inside the browser session, making the collection process faster and more stable than a purely DOM-based method.

The scraper retrieves both transcript metadata (company name, event date, sector, industry, transcript ID) and full transcript content (speaker names, roles, spoken text). Because the search endpoint may fail to return all results for very large date ranges, the scraper automatically splits large periods into smaller sub-periods and repeats the process recursively until the result size becomes manageable. Transcripts are downloaded in small batches and can be processed in parallel across monthly chunks. The pipeline includes retry mechanisms, automatic re-authentication, and checkpoint files to resume collection after interruptions.

Each transcript is saved as a structured text file organised by year, with a metadata header followed by chronological speaker blocks. After collection, transcripts are parsed to extract metadata, speaker turns, and a separation between *prepared remarks* and the *Q&A session*. Management text is built by concatenating all executive speaker blocks. Company names are mapped to ticker symbols through a three-step procedure: curated dictionary $\rightarrow$ exact match $\rightarrow$ fuzzy matching with rapidfuzz (threshold 75).

2.2 Market Data

Daily stock prices, market capitalisation, and earnings-surprise data are obtained from course-provided datasets. From daily closing prices we compute log returns $r_t = \ln(P_t/P_{t-1})$ and derive: (i) **post-call volatility** — annualised realised volatility over $[+1, +5]$ trading days, $\sigma_{\text{post}} = \text{std}(r_t) \times \sqrt{252}$; (ii) **pre-call volatility** over $[-30, -6]$ days; (iii) **abnormal volatility** $= \sigma_{\text{post}} - \sigma_{\text{pre}}$; (iv) **log market cap**; and (v) **earnings surprise**.

2.3 Sample Construction

The final analysis panel is constructed by merging transcript-level uncertainty scores with volatility measures and control variables. After requiring non-missing values for the uncertainty score, post-call volatility, pre-call volatility, log market capitalisation, and earnings surprise, the regression sample comprises **10,629 transcript–event observations** covering several hundred unique firms across 11 GICS sectors and 20 year-quarter periods.

3 Methodology

3.1 Labeling Strategy

Training a supervised text classifier requires labelled data. Because manually annotating tens of thousands of sentences for uncertainty is impractical, we adopt a *silver-labeling* approach using the Loughran and McDonald (2011) Uncertainty word list — a curated dictionary of terms (e.g., “approximate”, “contingency”, “uncertain”, “volatility”) known to signal uncertainty in financial disclosures.

For each sentence in our corpus, we compute an uncertainty ratio:

$$\text{uncertainty_ratio} = \frac{\text{count of uncertainty words}}{\text{total tokens in sentence}}.$$

Sentences with 5 to 200 tokens are retained. Among the subset of sentences with at least one uncertainty word, we assign binary labels based on the distribution of this ratio: sentences in the *top quartile* are labelled as “high uncertainty” ($y = 1$), while those in the *bottom quartile* are labelled as “low uncertainty” ($y = 0$). The middle two quartiles are discarded to create a clear separation between classes.

We then sample 15,000 sentences per class (30,000 total) and split them using stratified sampling: 80% training, 10% validation, 10% test. This lexicon-based silver-labeling strategy is scalable and reproducible, while the subsequent model training allows the classifier to learn contextual patterns that go beyond simple keyword presence.

3.2 Model Architecture and Training

We fine-tune **FinBERT** (Araci, 2019), a BERT-base architecture (12 transformer layers, 768 hidden dimensions, 12 attention heads, ~ 110 M parameters) pre-trained on a large financial-text corpus by ProsusAI (2023). A binary classification head is added on top of the [CLS] token embedding.

Table 1: Training hyperparameters.

Parameter	Value
Learning rate	2×10^{-5}
Batch size	32
Max sequence length	128 tokens
Epochs	4
Warmup ratio	0.1
Weight decay	0.01
Precision	FP16 (mixed)
Early stopping	patience 2 (by F1)

The model is trained using the Hugging Face Trainer API with the hyperparameters in Table 1. The best checkpoint, selected by validation F1, achieves the following performance on the held-out test set:

Table 2: Test-set evaluation metrics.

	Accuracy	F1	Precision	Recall
FinBERT (fine-tuned)	0.813	0.819	0.791	0.850

An F1 of 0.82 indicates that the model reliably distinguishes high-uncertainty from low-uncertainty sentences, with balanced precision–recall trade-off.

3.3 Transcript-Level Uncertainty Scoring

After training, we apply the classifier to every sentence in all 100,000+ transcripts. Each sentence receives a predicted probability $\hat{p}_i \in [0, 1]$ of being “high uncertainty.” We aggregate these sentence-level probabilities into two transcript-level measures:

- **Uncertainty score:** the mean predicted probability across all sentences in the transcript, $\bar{p} = \frac{1}{N} \sum_{i=1}^N \hat{p}_i$.
- **Uncertainty share:** the fraction of sentences classified as uncertain (i.e., $\hat{p}_i > 0.5$).

Both measures are also computed separately for the prepared-remarks and Q&A sections, enabling section-level analysis in robustness checks.

Scoring 130,000 transcripts with a transformer model poses a significant computational challenge: on a local CPU, inference alone would take an estimated 250 days. To address this, we convert all parsed transcripts into compact Parquet files and run both model training and full-corpus inference on a Google Colab Pro instance equipped with an NVIDIA A100 GPU, reducing training to ~ 50 minutes and scoring to ~ 6 hours.

3.4 Econometric Specification

Our main regression model is estimated using PanelOLS:

$$\begin{aligned} \text{PostCallVol}_{it} = & \beta_1 \text{UncertaintyScore}_{it} + \beta_2 \text{PreCallVol}_{it} \\ & + \beta_3 \ln(\text{MarketCap})_{it} + \beta_4 \text{EarnSurprise}_{it} \\ & + \alpha_s + \gamma_q + \varepsilon_{it} \end{aligned} \quad (1)$$

where i denotes the firm, t the earnings-call event, α_s are sector fixed effects, and γ_q are year-quarter fixed effects. Standard errors are clustered at the ticker level to account for within-firm serial correlation.

All continuous variables are winsorised at the 1st and 99th percentiles. To facilitate economic interpretation, the uncertainty score is scaled by 100 (expressed in percentage points) and volatility by 10,000 (expressed in basis points).

We estimate four specifications:

1. **Baseline:** uncertainty score alone, no controls or fixed effects.
2. **Full controls:** all controls and fixed effects (Equation 1).
3. **Share measure:** replaces the uncertainty score with the uncertainty share as the key independent variable.
4. **Abnormal volatility:** uses abnormal volatility ($\sigma_{\text{post}} - \sigma_{\text{pre}}$) as the dependent variable.

This design is free from look-ahead bias: all control variables are observed on or before the call date, while the dependent variable is computed exclusively from post-call returns.

4 Results

4.1 Descriptive Evidence

Both uncertainty measures are right-skewed, centred around 10%, with a long right tail above 30%, confirming meaningful cross-transcript variation (Figure 7, Appendix C). Quarterly trends show a peak in late 2022 / early 2023 — consistent with heightened macroeconomic uncertainty — followed by a gradual decline (Figure 8). Across sectors, Energy, Utilities, and Real Estate exhibit the highest uncertainty, while Communication Services and Consumer Staples rank lowest (Figure 9).

4.2 Main Regression Results

Table 3 reports the results of the four regression specifications described in Section 3.4.

Table 3: Panel regression results. The dependent variable is annualised post-call realised volatility in basis points, except in Model 4 where it is abnormal volatility. The uncertainty measure is scaled to percentage points. Standard errors (in parentheses) are clustered by ticker. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	(1) Baseline	(2) Full	(3) Share	(4) Abnormal Vol
Uncertainty Score (pp)	26.26 (20.93)	49.06*** (16.73)		47.82*** (12.71)
Uncertainty Share (pp)			48.75*** (16.70)	
Pre-Call Volatility		5,329.89*** (370.24)	5,329.69*** (370.22)	
Log Market Cap		−367.15*** (39.65)	−367.22*** (39.65)	
Earnings Surprise		7.88 (30.17)	7.87 (30.17)	
Sector FE	No	Yes	Yes	No
Year-Quarter FE	No	Yes	Yes	No
Clustered SE	Yes	Yes	Yes	Yes
R^2	0.008	0.255	0.255	0.020
N	18,206	10,629	10,629	17,079

Without controls (column 1), the uncertainty coefficient is positive but insignificant ($\hat{\beta}_1 = 26.26$, $p = 0.210$), due to omitted cross-sectional confounders. Adding pre-call volatility, firm size, earnings surprise, and fixed effects (column 2) nearly doubles the coefficient: $\hat{\beta}_1 = 49.06$ ($p = 0.003$). **A 1 percentage point increase in model-predicted uncertainty is associated with approximately 49 basis points higher annualised post-call realised volatility.** Given an interquartile range of roughly 5 percentage points, this implies a difference of about 245 bps between the 75th and 25th percentile transcripts.

The controls behave as expected: pre-call volatility is strongly positive (5,330 bps, volatility persistence), log market cap is strongly negative (−367 bps, larger firms are less volatile), and earnings surprise is insignificant ($p = 0.794$). Replacing the uncertainty score with the uncertainty share (column 3) or using abnormal volatility as the dependent variable (column 4) yields nearly identical coefficients (48.75 and 47.82 bps, both $p < 0.005$), confirming robustness to the choice of measure and dependent variable. The correlation matrix (Figure 10, Appendix C) confirms that no regressor pair exceeds $r = 0.40$, ruling out multicollinearity concerns.

4.3 Robustness Checks

We conduct five sets of robustness tests, summarised in Table 4 and in the forest plot in Figure 11 (Appendix C).

Table 4: Robustness checks. All specifications include the full set of controls and fixed effects from Model 2 unless otherwise noted. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Test	Key Variable	Coeff.	(SE)	p	N
<i>(a) Alternative volatility windows</i>					
Vol [+1, +5] (main)	Model Uncertainty	49.06***	(16.73)	0.003	10,629
Vol [+1, +3]	Model Uncertainty	61.03***	(22.64)	0.007	10,629
Vol [+1, +10]	Model Uncertainty	27.53**	(13.98)	0.049	10,629
Abnormal volatility	Model Uncertainty	48.23***	(15.97)	0.003	10,629
<i>(b) Lexicon-only baseline</i>					
Vol [+1, +5]	Lexicon Uncertainty	445.41*	(268.95)	0.098	10,629
<i>(c) Section split</i>					
Prepared remarks	Prep. Remarks Unc.	89.91***	(21.69)	<0.001	2,422
Q&A session	Q&A Uncertainty	15.84	(15.26)	0.299	10,156
<i>(d) Firm size</i>					
Large firms	Model Uncertainty	44.62***	(16.92)	0.008	6,423
Small firms	Model Uncertainty	49.46*	(27.89)	0.076	4,206
<i>(e) Sector heterogeneity</i>					
Health Care	Model Uncertainty	112.26*	(58.14)	0.054	1,169
Financials	Model Uncertainty	64.72*	(38.92)	0.096	1,200
Industrials	Model Uncertainty	48.66	(38.16)	0.202	1,906
Inf. Technology	Model Uncertainty	19.30	(42.84)	0.652	2,180
Cons. Discretionary	Model Uncertainty	-22.36	(45.52)	0.623	1,489

(a) Alternative windows. The effect is strongest over [+1, +3] (61.03 bps, $p = 0.007$) and attenuates over [+1, +10] (27.53 bps, $p = 0.049$), consistent with gradual information absorption. Abnormal volatility yields 48.23 bps ($p = 0.003$).

(b) Lexicon-only baseline. Replacing the FinBERT score with a raw Loughran–McDonald keyword ratio produces a large but imprecise coefficient (445.41, $p = 0.098$), confirming that the fine-tuned model captures uncertainty more precisely than dictionary counts.

(c) Section split. Prepared-remarks uncertainty is strongly predictive (89.91 bps, $p < 0.001$), while Q&A uncertainty is not (15.84 bps, $p = 0.299$). When management is uncertain in their scripted remarks, the market treats it as especially informative.

(d) Firm size & (e) Sectors. The effect holds for both large (44.62 bps, $p = 0.008$) and small firms (49.46 bps, $p = 0.076$). It is strongest in Health Care (112.26 bps) and Financials (64.72 bps) — sectors with salient regulatory and credit uncertainty — and weak or negative in Information Technology and Consumer Discretionary.

5 Discussion and Conclusion

This paper demonstrates a statistically significant and economically meaningful link between management uncertainty language in earnings calls and short-term stock volatility. A 1 percentage point increase in the model-predicted uncertainty score is associated with approximately 49 basis points higher annualised realised volatility over the five trading days following

the call.

Several insights emerge from our analysis. First, the distinction between prepared remarks and Q&A is crucial: the entire predictive content of uncertainty is concentrated in the scripted portion of the call. This is consistent with the idea that management carefully crafts prepared remarks, so uncertainty language in this section reflects genuine apprehension about the firm’s outlook, whereas Q&A responses are more spontaneous and noisy.

Second, the sector heterogeneity reveals that the uncertainty–volatility relationship is strongest in industries characterised by complex regulation and high-stakes uncertainty (Health Care, Financials) and weakest in sectors with high baseline idiosyncratic volatility (Information Technology).

Third, comparing the fine-tuned FinBERT model with a pure lexicon baseline confirms the value of contextual language models: the FinBERT-based measure produces tighter standard errors and more significant coefficients, demonstrating that a supervised model trained on domain-specific silver labels outperforms raw dictionary counts.

Limitations. Our study has several limitations that should be noted. First, the silver labels derived from the Loughran–McDonald lexicon may fail to capture more nuanced forms of uncertainty (e.g., hedged optimism, implicit uncertainty through conditional phrasing). Second, there is an inherent endogeneity concern: firms facing genuinely uncertain futures may simultaneously use uncertain language *and* experience volatile stock prices, making it difficult to establish a causal direction. We control for pre-call volatility and firm size to mitigate this, but a causal-identification strategy (e.g., instrumental variables or a natural experiment) remains desirable for future work. Third, our sample is limited to firms for which we could successfully match transcripts to market data, which may introduce a selection bias toward larger, more liquid firms.

Extensions. Future work could explore instrumental variable approaches for causal identification, multi-class uncertainty categorisation (distinguishing risk uncertainty from opportunity uncertainty), and interactions between textual uncertainty and macroeconomic conditions such as monetary-policy regime changes.

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A Pipeline Architecture

The full analytical pipeline consists of 12 sequential steps, implemented in modular Python scripts orchestrated by `run_pipeline.py`:

1. **Parse Transcripts** — Convert $\sim 100\text{K}$ raw .txt files into a structured Parquet dataset with metadata, speaker blocks, and section splits.
2. **Map Tickers** — Fuzzy-match company names to ticker symbols using a three-tier approach (curated dictionary $\rightarrow$ exact match $\rightarrow$ fuzzy match at score ≥ 75).
3. **Lexicon Scoring** — Score every sentence using the Loughran–McDonald Uncertainty word list; compute per-sentence uncertainty ratios.
4. **Build Training Set** — Select top/bottom quartile sentences, balance classes at 15K each, and split 80/10/10.
5. **Train FinBERT** — Fine-tune ProsusAI/finbert for binary uncertainty classification (GPU-accelerated, ~ 10 min on A100).
6. **Evaluate Model** — Compute test-set accuracy, F1, precision, recall, and confusion matrix.
7. **Predict Uncertainty** — Score all transcripts with the fine-tuned model; compute per-transcript uncertainty score and share.
8. **Compute Volatility** — Calculate post-call, pre-call, and abnormal realised volatility from daily log returns.
9. **Build Panel** — Merge uncertainty scores, volatility, market cap, and earnings surprise into the final panel dataset.
10. **Regression Analysis** — Estimate four PanelOLS specifications with sector and time fixed effects.
11. **Robustness Checks** — Alternative windows, lexicon-only, section splits, size splits, sector sub-samples.
12. **Visualisations** — Generate all figures for the report.

All intermediate outputs are stored as Parquet files in `data/processed/` for efficient I/O. The pipeline supports modular execution (`--from`, `--only`, `--skip-training`) for reproducibility and convenience.

B Training History

Table 5 shows the evolution of validation metrics across training epochs.

Table 5: Training history: validation metrics per epoch.

Epoch	Val Loss	F1	Accuracy	Precision / Recall
1	0.602	0.721	0.725	0.741 / 0.700
2	0.515	0.760	0.763	0.771 / 0.750
3	0.487	0.769	0.775	0.790 / 0.750
4	0.480	0.819	0.813	0.791 / 0.850

The model improves steadily across all four epochs, with the largest gain in recall at epoch 4.

The early-stopping criterion (patience of 2 epochs without F1 improvement) was not triggered; the model was saved at the final epoch.

C Descriptive and Robustness Figures

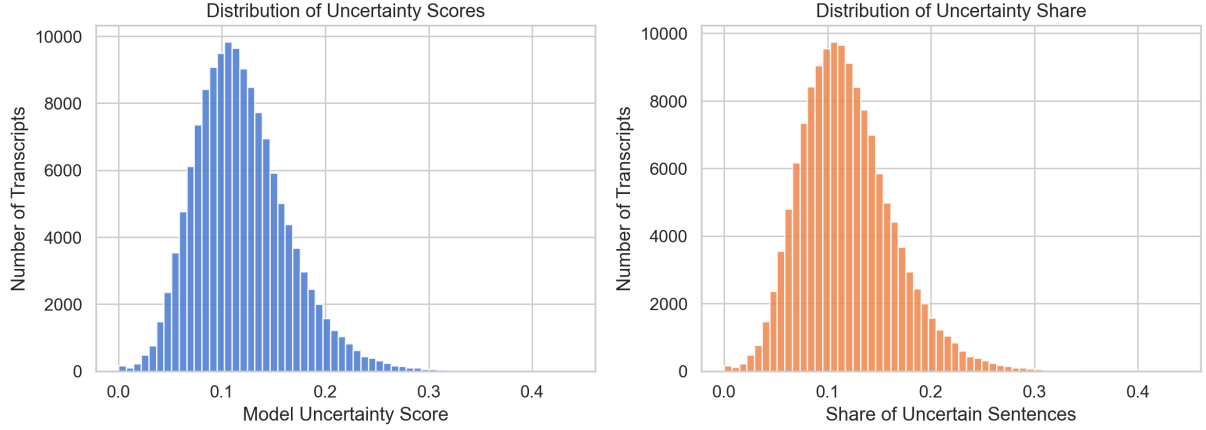


Figure 7: Distribution of transcript-level uncertainty measures. *Left*: mean predicted probability of high uncertainty. *Right*: share of sentences classified as uncertain ($\hat{p} > 0.5$).

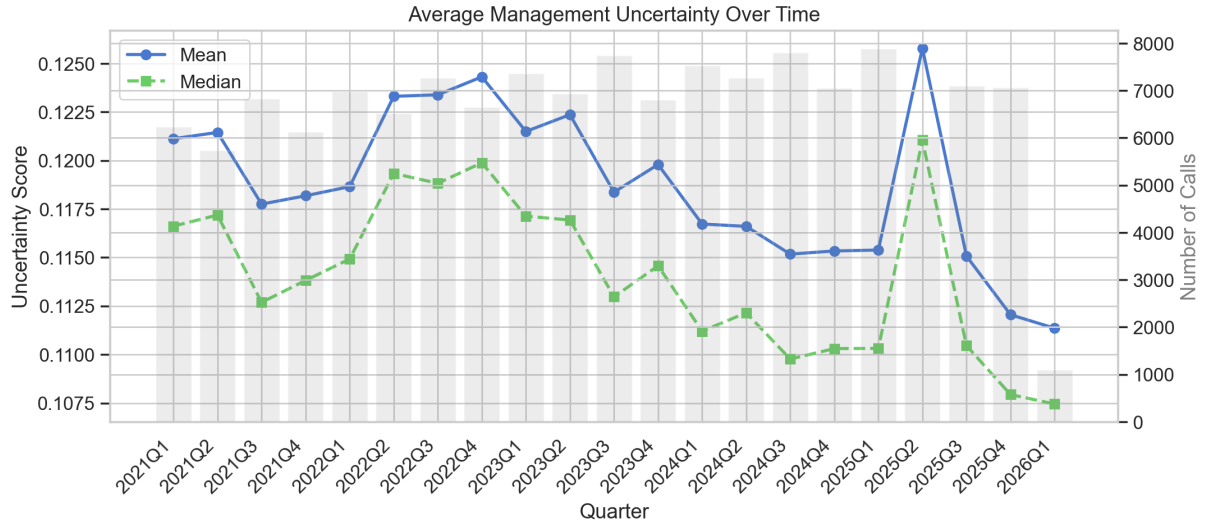


Figure 8: Average management uncertainty over time (quarterly). Blue solid line: mean; green dashed line: median; grey bars: number of calls.

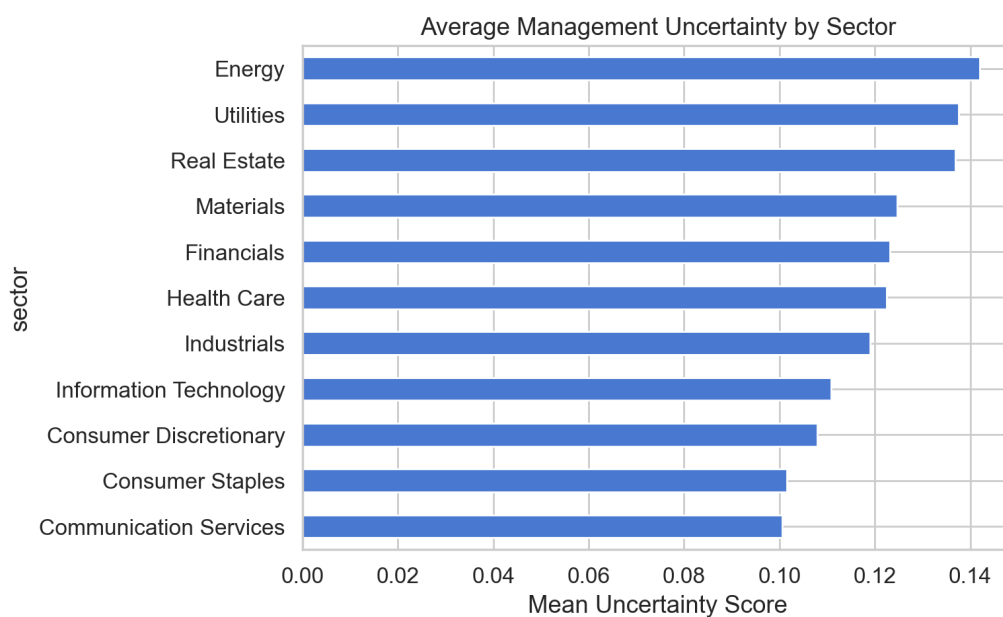


Figure 9: Average management uncertainty score by GICS sector.

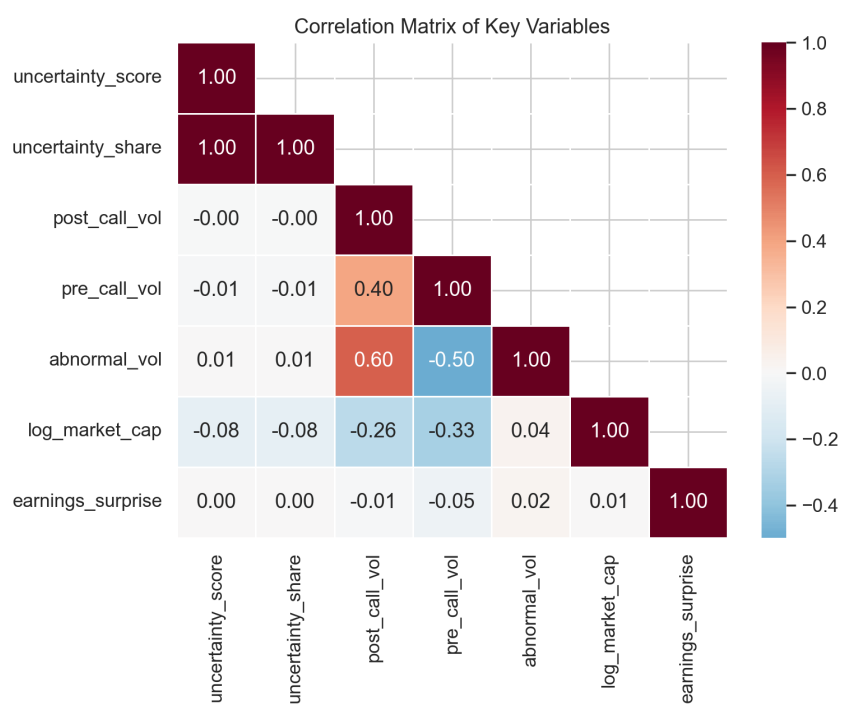


Figure 10: Correlation matrix of key panel variables.

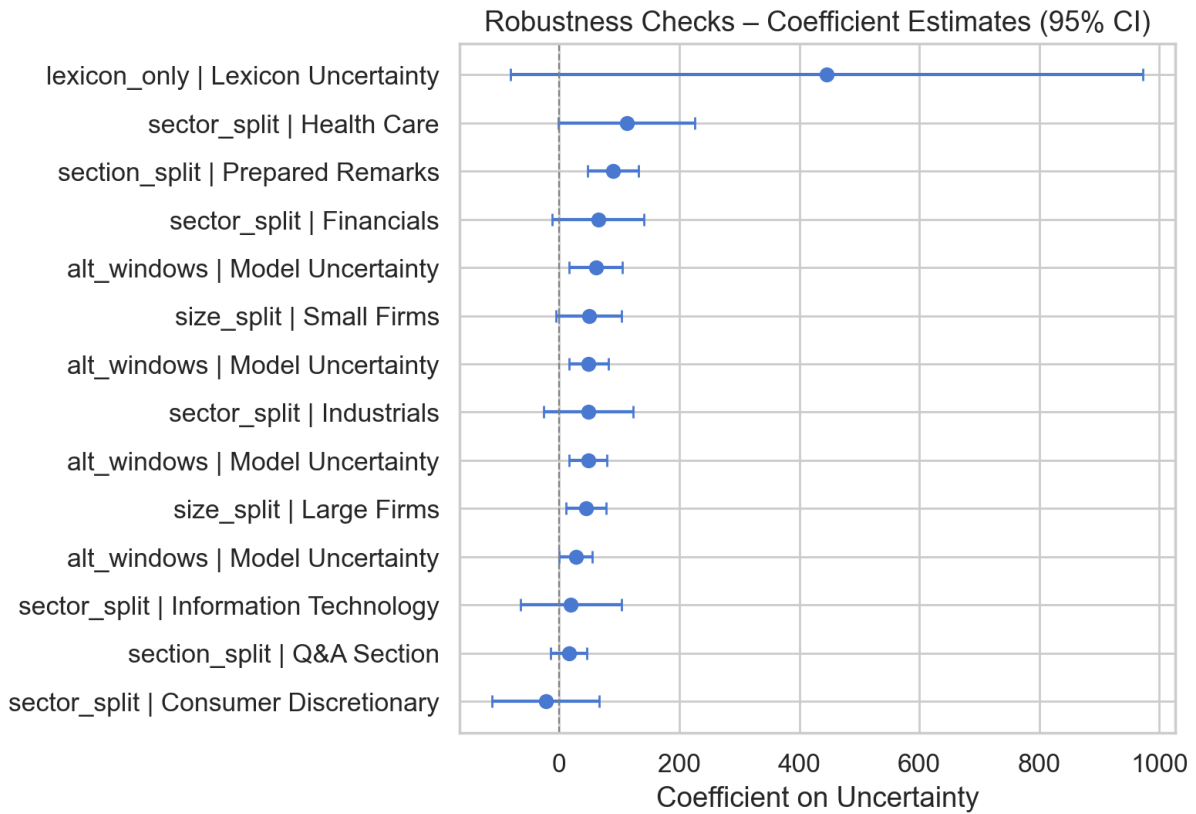


Figure 11: Forest plot of uncertainty coefficients across all robustness specifications. Points: estimated coefficient; bars: 95% CI; dashed line: zero.

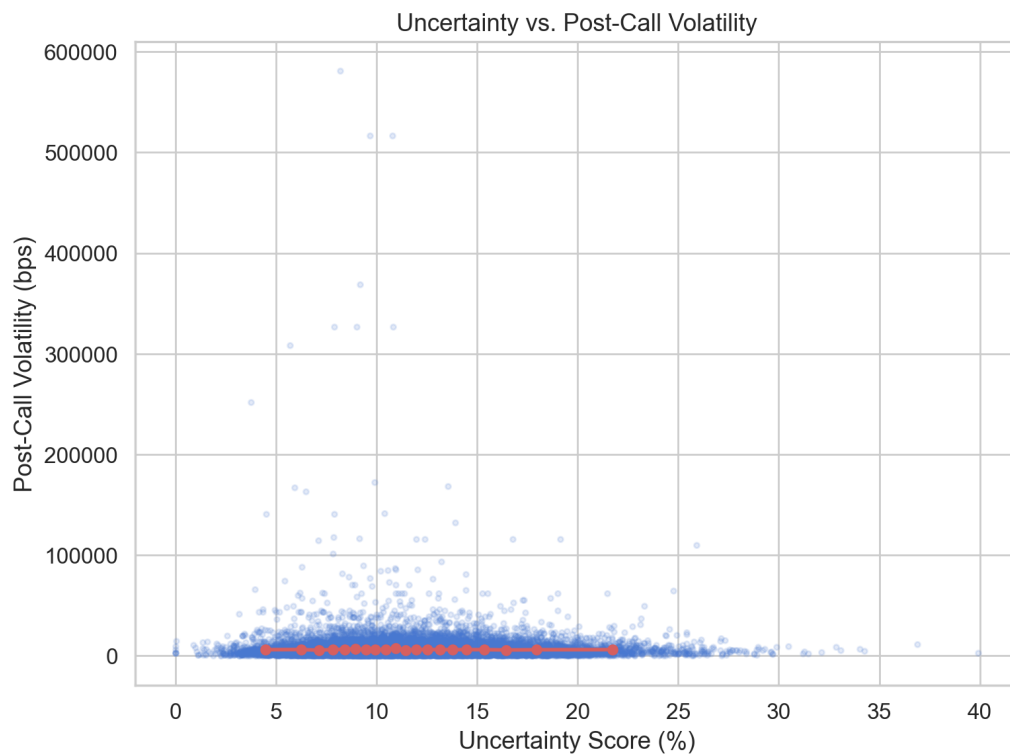


Figure 12: Scatter plot of uncertainty score (%) versus post-call realised volatility (bps). Red circles show binned means.